

Use EBI-(P/G)DMA to drive RGB Sync-type LCM

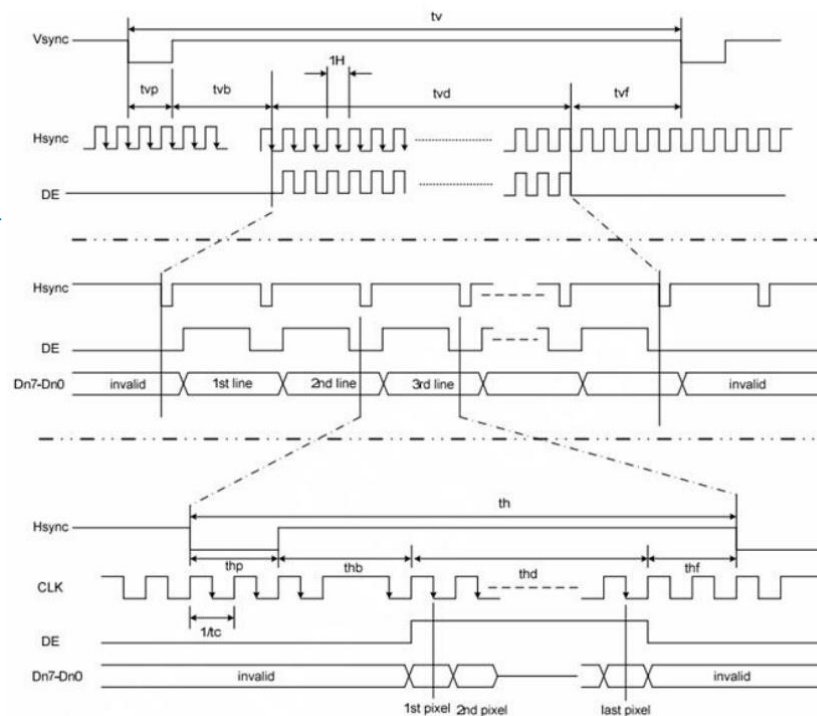
For NuMicro MCU-family HMI Solution

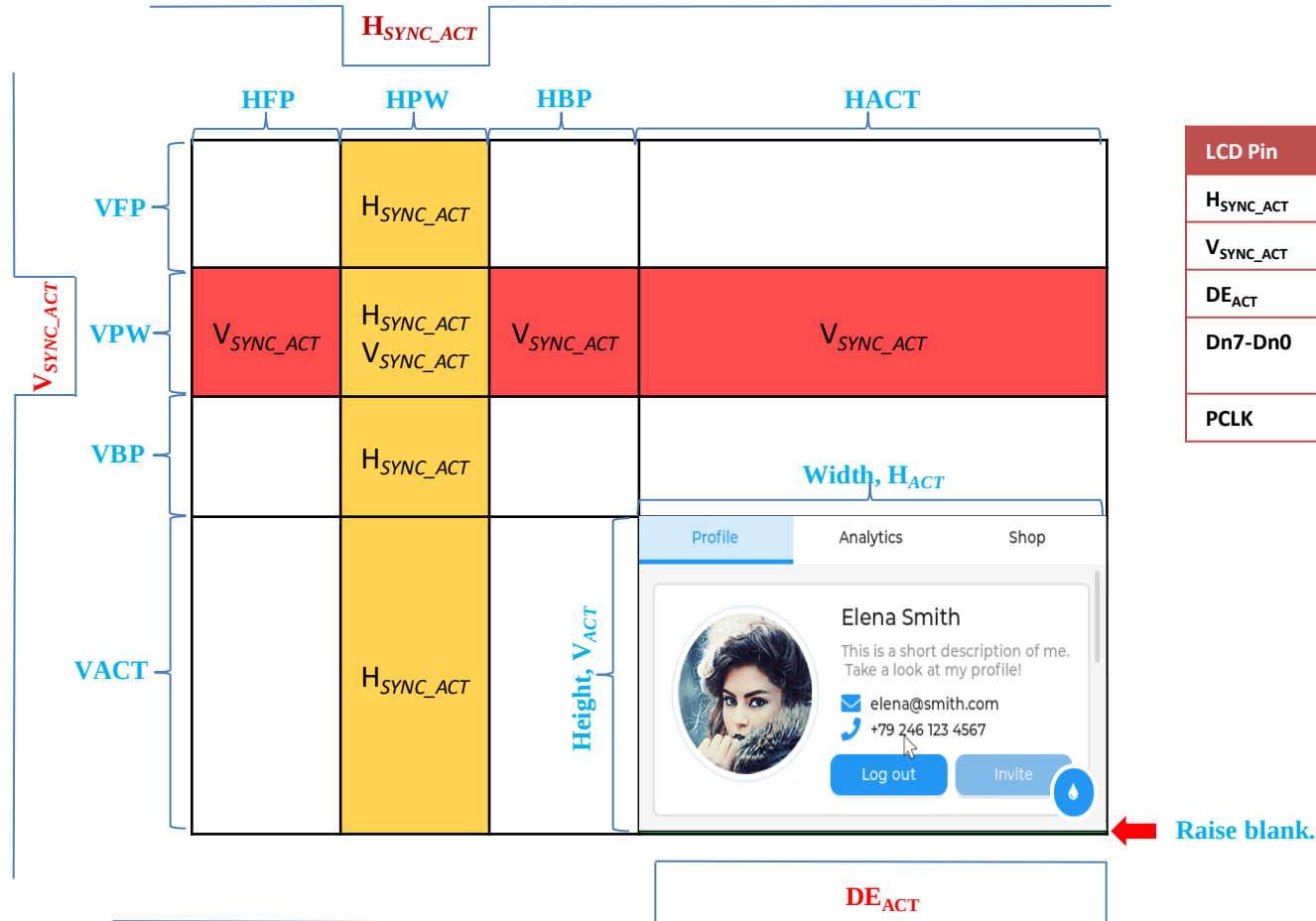
An Example – RGB Sync-type LCD Timing

- **Use EBI-DMA to simulate a software VPOST.**
 - Traversal V*H DMA descriptors for post a frame.
 - Memory-to-memory transferring.
 - Raising an interrupt event when latest DMA transaction in descriptor list is done.
 - Support to switch new VRAM address in Interrupt Service Routine to avoid tearing issue.
 - Outputted FPS depends on EBI-bus clock frequency.

- **Signal pins.**

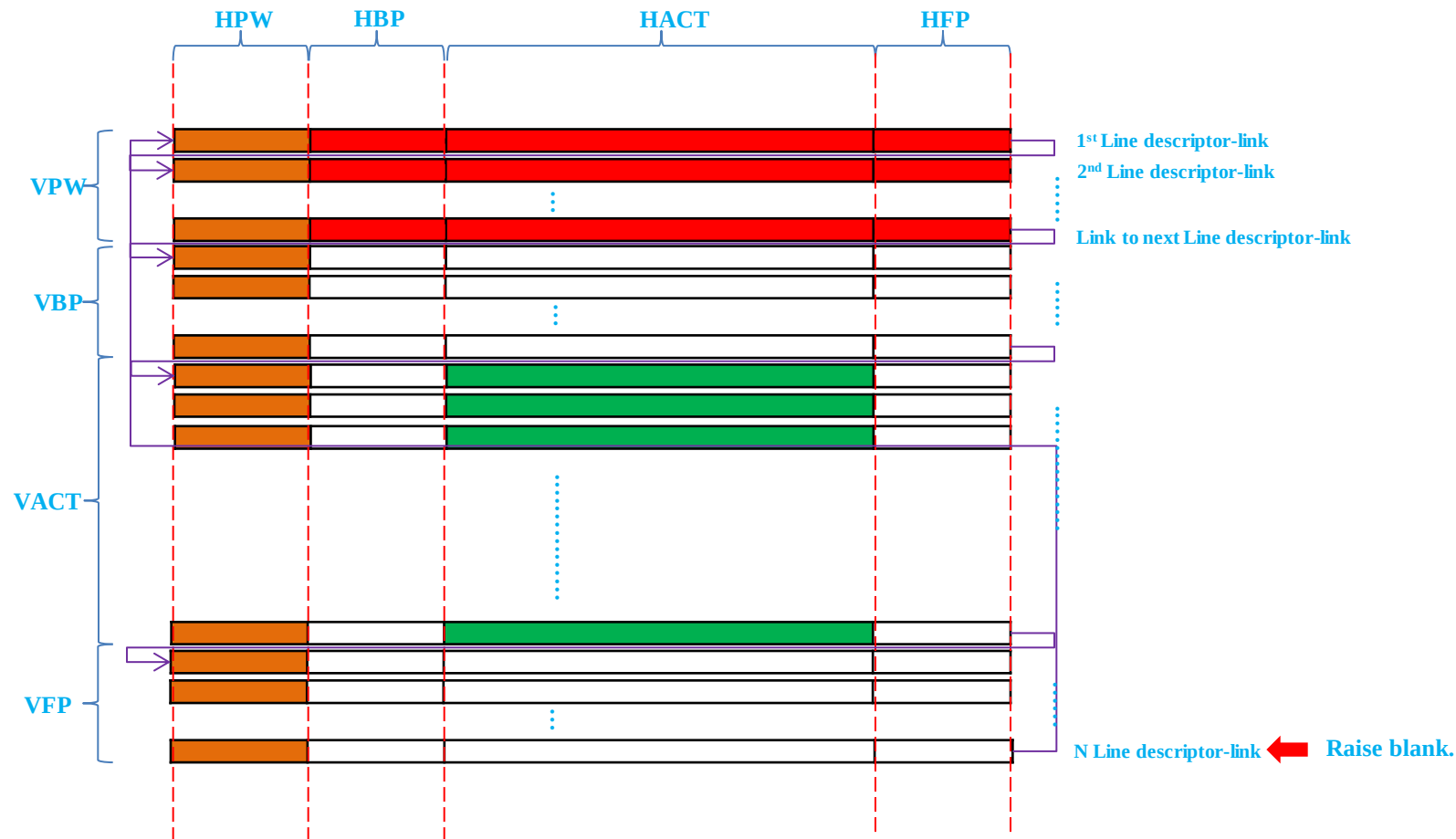
- Use 3-EBI ADR pins to act Vsync / Hsync / DE
- Use EBI nWR to act pixel clock signal pin
- Use 16-EBI AD pin to act RGB565 signal pins

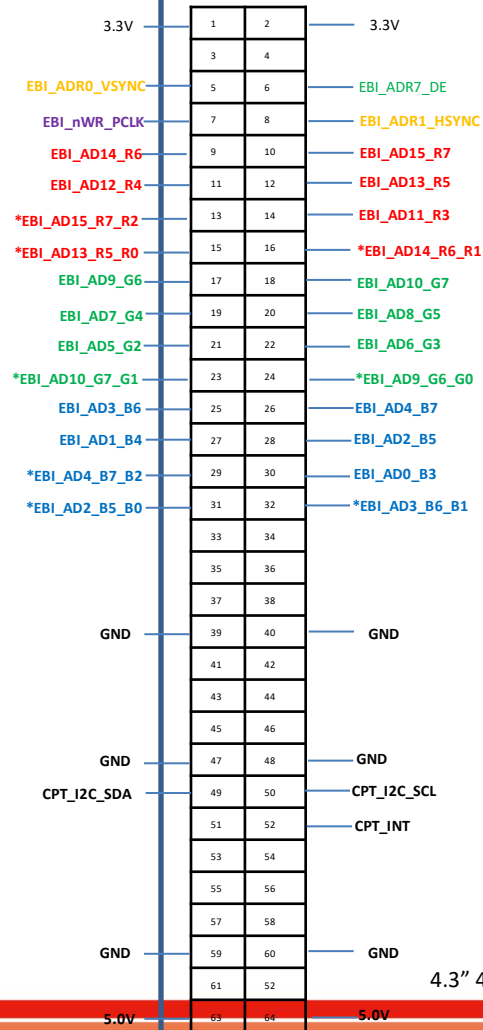
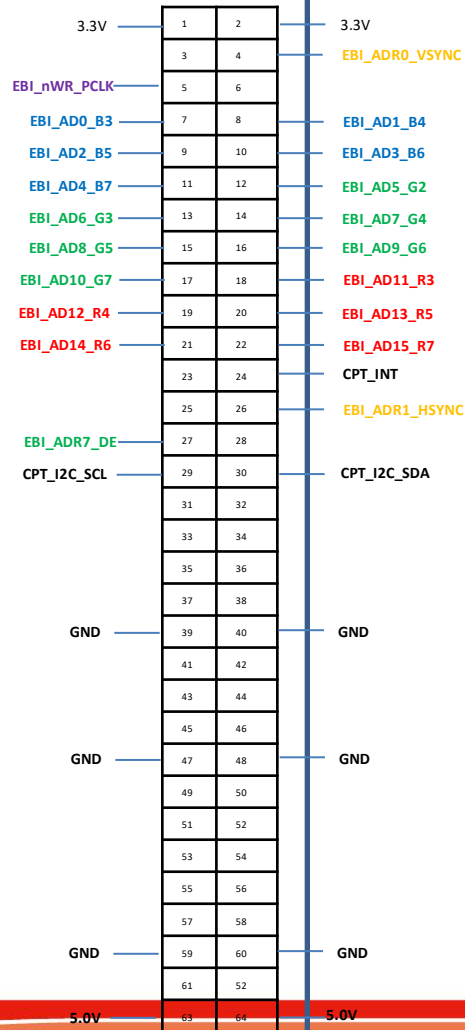




LCD Pin	EBI Pin	EBI Write (16b)
H_SYNC_ACT	EBI_ADR1_PH6	Address: 0x00000004
V_SYNC_ACT	EBI_ADR0_PH7	Address: 0x00000002
DE_ACT	EBI_ADR7_PH0	Address: 0x00000100
Dn7-Dn0	EBI_AD[0-15]	Dummy(Zero) Or RGB565 Pixel data
PCLK	EBI_nWR_PJ9	Every EBI-Writing

Stage	Name
HPW	Horizontal Pulse Width
HBP	Horizontal Back Porch
HACT	Horizontal Active
HFP	Horizontal Front Porch
VPW	Vertical Pulse Width
VBP	Vertical Back Porch
VACT	Vertical Active
VFP	Vertical Front Porch





Driver definitions

```

#define CONFIG_TIMING_HACT      480 /*!< XRES */
#define CONFIG_TIMING_VACT      272 /*!< YRES */
#define CONFIG_TIMING_HBP       30 /*!< HBP (Horizontal Back Porch) */
#define CONFIG_TIMING_HFP       5  /*!< HFP (Horizontal Front Porch) */
#define CONFIG_TIMING_HPWW      41 /*!< HPW (HSYNC plus width) */
#define CONFIG_TIMING_VBP       2  /*!< VBP (Vertical Back Porch) */
#define CONFIG_TIMING_VFP       27 /*!< VFP (Vertical Front Porch) */
#define CONFIG_TIMING_VPW       10 /*!< VPW (VSYNC width) */

#define CONFIG_DISP_VPW_ACTIVE_LOW 1 /*!< Enable VPW active low */
#define CONFIG_DISP_HPWW_ACTIVE_LOW 1 /*!< Enable HPW active low */
#define CONFIG_DISP_DE_ACTIVE_LOW 0 /*!< Disable DE active low */

#define CONFIG_DISP_VSYNC_BITIDX 1 /*!< SET_EBI_ADR0_PH7 */
#define CONFIG_DISP_HSYNC_BITIDX 2 /*!< SET_EBI_ADR1_PH6 */
#define CONFIG_DISP_DE_BITIDX 8 /*!< SET_EBI_ADR7_PH0 */

/* Don't touch me */
#define CONFIG_DISP_VSYNC_BITMASK (1<<CONFIG_DISP_VSYNC_BITIDX)
#define CONFIG_DISP_HSYNC_BITMASK (1<<CONFIG_DISP_HSYNC_BITIDX)
#define CONFIG_DISP_DE_BITMASK (1<<CONFIG_DISP_DE_BITIDX)
#define CONFIG_DISP_VPW_ACTIVE_LOW?(-1*CONFIG_DISP_VSYNC_BITMASK):CONFIG_DISP_VSYNC_BITMASK)
#define CONFIG_DISP_HPWW_ACTIVE_LOW?(-1*CONFIG_DISP_HSYNC_BITMASK):CONFIG_DISP_HSYNC_BITMASK)
#define CONFIG_DISP_DE_ACTIVE_LOW?(-1*CONFIG_DISP_DE_BITMASK):CONFIG_DISP_DE_BITMASK)
#define CONFIG_DISP_EBI_ADDR (EBI_BANK0_BASE_ADDR+(CONFIG_DISP_EBI*EBI_MAX_SIZE)+\
(CONFIG_DISP_VPW_ACTIVE_LOW<<CONFIG_DISP_VSYNC_BITIDX)+\
(CONFIG_DISP_HPWW_ACTIVE_LOW<<CONFIG_DISP_HSYNC_BITIDX)+\
(CONFIG_DISP_DE_ACTIVE_LOW<<CONFIG_DISP_DE_BITIDX))

```

Driver Enumeration/Structure

```
typedef enum
{
    evHStageHFP,
    evHStageHSYNC,
    evHStageHBP,
    evHStageHACT,
    evHStageCNT
} E_HSTAGE;

typedef enum
{
    evVStageVFP,
    evVStageVSYNC,
    evVStageVBP,
    evVStageVACT,
    evVStageCNT
} E_VSTAGE;

typedef struct
{
    DSCT_T    m_dsch[evHStageCNT];
} S_DSC_HLINE;

typedef struct
{
    #define DEF_TOTAL_VLINES (CONFIG_TIMING_VPW+CONFIG_TIMING_VBP+CONFIG_TIMING_VACT+CONFIG_TIMING_VFP)
    S_DSC_HLINE m_dscV[DEF_TOTAL_VLINES];
} S_DSC_LCD;
```